SQL JOINs

05 · Joins — Complete Reference Guide

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1. Basic Joins

→ No Join — retrieve customers and orders as separate results

```
SELECT * FROM customers;
SELECT * FROM orders;
```

→ INNER JOIN — only customers who have placed an order

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.sales
FROM customers AS c
INNER JOIN orders AS o
    ON c.id = o.customer_id
```

→ LEFT JOIN — all customers, including those without orders

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.sales
FROM customers AS c
LEFT JOIN orders AS o
    ON c.id = o.customer_id
```

→ RIGHT JOIN — all orders, including those without matching customers

```
SELECT
```

```
    c.id,
    c.first_name,
    o.order_id,
    o.customer_id,
    o.sales
FROM customers AS c
RIGHT JOIN orders AS o
    ON c.id = o.customer_id
```

→ Alternative to RIGHT JOIN — rewrite using LEFT JOIN

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.sales
FROM orders AS o
LEFT JOIN customers AS c
    ON c.id = o.customer_id
```

→ FULL JOIN — all customers and all orders, even if no match

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.customer_id,
    o.sales
FROM customers AS c
FULL JOIN orders AS o
    ON c.id = o.customer_id
```

2. Advanced Joins

→ LEFT ANTI JOIN — customers who have NOT placed any order

```
SELECT *
FROM customers AS c
LEFT JOIN orders AS o
    ON c.id = o.customer_id
WHERE o.customer_id IS NULL
```

→ RIGHT ANTI JOIN — orders with no matching customers

```
SELECT *
FROM customers AS c
RIGHT JOIN orders AS o
    ON c.id = o.customer_id
WHERE c.id IS NULL
```

→ Alternative to RIGHT ANTI JOIN using LEFT JOIN

```
SELECT *
FROM orders AS o
LEFT JOIN customers AS c
    ON c.id = o.customer_id
WHERE c.id IS NULL
```

→ Alternative INNER JOIN using LEFT JOIN + WHERE

```
SELECT *
FROM customers AS c
LEFT JOIN orders AS o
    ON c.id = o.customer_id
WHERE o.customer_id IS NOT NULL
```

→ FULL ANTI JOIN — customers without orders AND orders without customers

```
SELECT
    c.id,
    c.first_name,
    o.order_id,
    o.customer_id,
    o.sales
FROM customers AS c
FULL JOIN orders AS o
    ON c.id = o.customer_id
WHERE o.customer_id IS NULL OR c.id IS NULL
```

→ CROSS JOIN — all possible combinations of customers and orders

```
SELECT *
FROM customers
CROSS JOIN orders
```

3. Multiple Table Joins (4 Tables)

Task: Using SalesDB, retrieve a list of all orders with related customer, product, and employee details. Display: Order ID, Customer name, Product name, Sales amount, Product price, Salesperson name.

```
USE SalesDB

SELECT
    o.OrderID,
    o.Sales,
    c.FirstName   AS CustomerFirstName,
    c.LastName    AS CustomerLastName,
    p.Product     AS ProductName,
    p.Price,
```

```
    e.FirstName  AS EmployeeFirstName,
    e.LastName   AS EmployeeLastName
FROM Sales.Orders AS o
LEFT JOIN Sales.Customers AS c
    ON o.CustomerID  = c.CustomerID
LEFT JOIN Sales.Products AS p
    ON o.ProductID   = p.ProductID
LEFT JOIN Sales.Employees AS e
    ON o.SalesPersonID = e.EmployeeID
```